Solution: Conflicting Appointments

Problem Statement

Given an array of intervals representing 'N' appointments, find out if a person can **attend all the appointments**.

Example 1:

```
Appointments: [[1,4], [2,5], [7,9]]
Output: false
Explanation: Since [1,4] and [2,5] overlap, a person cannot attend both
of these appointments.
```

Example 2:

```
Appointments: [[6,7], [2,4], [8,12]]
Output: true
Explanation: None of the appointments overlap, therefore a person can
attend all of them.
```

Example 3:

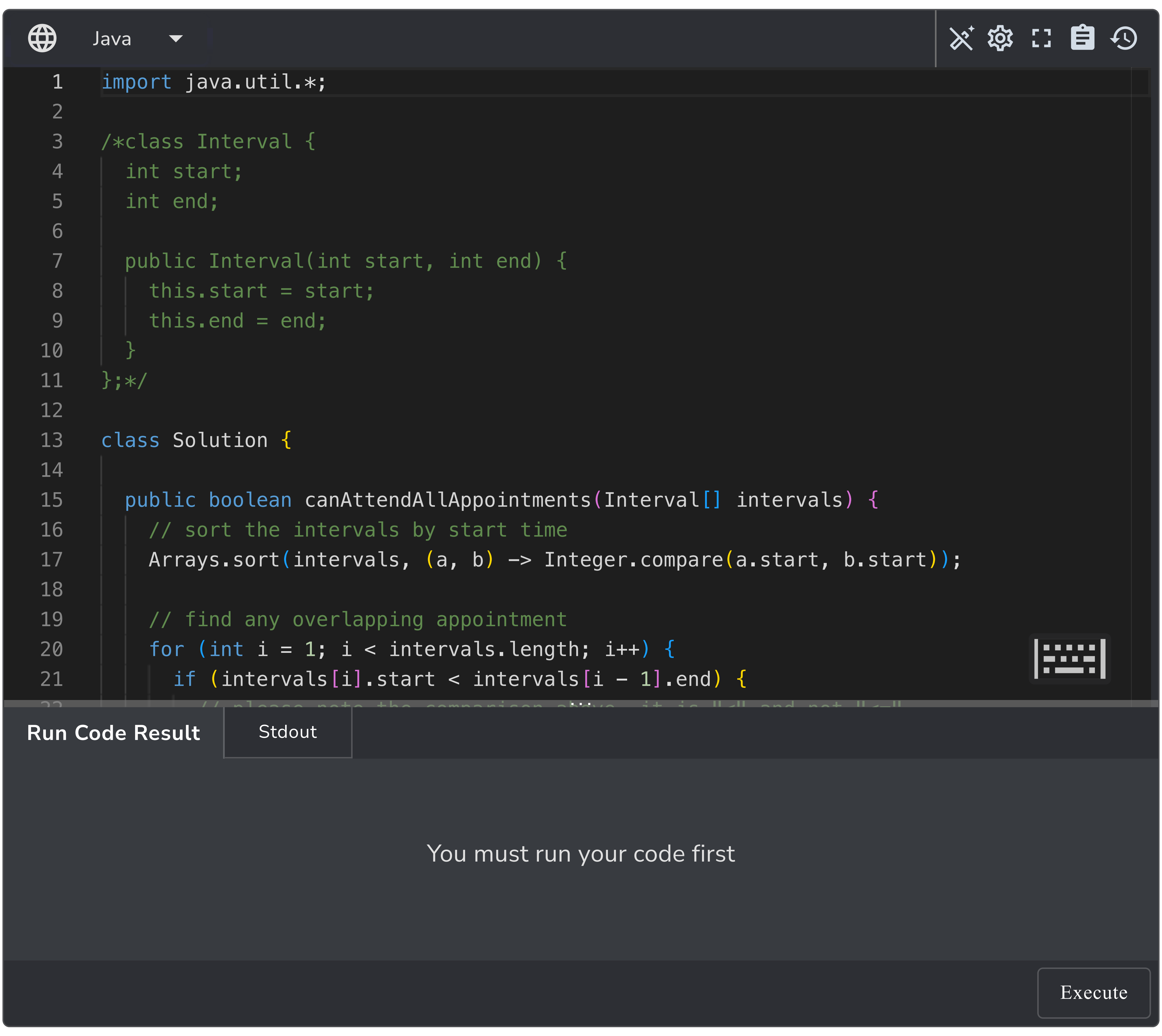
```
Appointments: [[4,5], [2,3], [3,6]]
Output: false
Explanation: Since [4,5] and [3,6] overlap, a person cannot attend both
of these appointments.
```

Solution

The problem follows the `Merge Intervals` pattern. We can sort all the intervals by start time and then check if any two intervals overlap. A person will not be able to attend all appointments if any two appointments overlap.

Code

Here is what our algorithm will look like:



```java
import java.util.*;

/*class Interval {
  int start;
  int end;

  public Interval(int start, int end) {
    this.start = start;
    this.end = end;
  }
};*/

class Solution {

  public boolean canAttendAllAppointments(Interval[] intervals) {
    // sort the intervals by start time
    Arrays.sort(intervals, (a, b) -> Integer.compare(a.start, b.start));

    // find any overlapping appointment
    for (int i = 1; i < intervals.length; i++) {
      if (intervals[i].start < intervals[i - 1].end) {
```

You must run your code first

Time Complexity

The time complexity of the above algorithm is O(N*logN), where 'N' is the total number of appointments. Though we are iterating the intervals only once, our algorithm will take O(N * logN) since we need to sort them in the beginning.

Space Complexity

The space complexity of the above algorithm will be O(N), which we need for sorting. For Java, Arrays.sort() uses Timsort, which needs O(N) space.

Similar Problems

Problem 1: Given a list of appointments, find all the conflicting appointments.

Example:

```
Appointments: [[4,5], [2,3], [3,6], [5,7], [7,8]]
Output:
[4,5] and [3,6] conflict.
[3,6] and [5,7] conflict.
```

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